WEAzer User Guide

Woofung Editing Analyzer (WEAzer)

Version: 1.0.0 beta

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Introduction

WEAzer is a powerful tool designed for researchers to manage, extract, analyze, and merge next-generation sequencing (NGS) data. This standalone Windows application provides an intuitive interface for handling Next Generation Sequencing sequence analysis. The main functions including barcode demultiplexing, editing analysis, and paired-end sequence merging. This guide covers the four main tabs of the application: Project Manager, Sequence Extractor, Editing Analyzer, and Paired-End Sequence Merger.

Before using WEAzer, ensure you agree to the License Agreement displayed upon first launch.

Project Manager

Overview

The **Project Manager** tab allows you to create, save, and load projects for your sequencing experiments. It enables you to define project details, including barcode sequences, anchor configurations, and primer sequences, which are essential for downstream analysis in WEAzer.

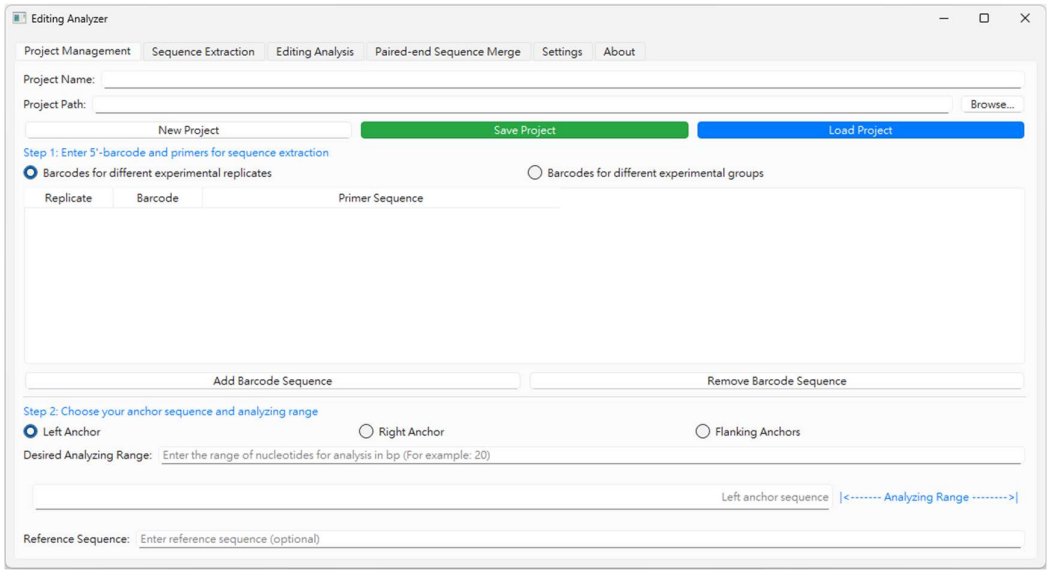
Features

- **Project Creation**: Define project name, path, and sequence settings.
- **Barcode Management**: Add and organize barcodes for replicates or experimental groups.
- **Anchor Configuration**: Set left, right, or flanking anchor sequences with an analysis range.
- **Reference Sequence**: Optionally specify a reference sequence for analysis.
- **Save/Load Projects**: Store project configurations as JSON files and reload them as needed.
- **Multilingual Support**: Switch between languages (e.g., English, Chinese, Portuguese, Korean) via the Settings tab.

How to Use

1. **Make a New Project**:
    - Click **New Project** to clear all fields and begin a new configuration. **Please note**: You will need to hit **Save Project** or **Load Project** bottom to apply the settings to other analyzing functions.
2. **Configure Barcodes (Optional)**:
    - choose **Barcodes for different experimental replicates** or **Barcodes for different experimental groups**.
    - Click **Add Barcode Sequence** to insert a row in the table.
    - Fill in the **Replicate/Group**, **Barcode**, and **Primer Sequence** columns (e.g., "Replicate 1", "GGAA", "GTGATTGCTGGCCGGAACA").
    - Use **Remove Barcode Sequence** to delete unwanted rows.
3. **Set Anchor Sequences (Optional)**:
    - Under **Step 2**, select an anchor type:
        - **Left Anchor**: Enter a sequence (e.g., "ATCG") and specify a range in base pairs (bp) (e.g., "20").

        - **Right Anchor**: Enter a sequence and range similarly.
        - **Flanking Anchors**: Enter both **Left Anchor** and **Right Anchor** sequences (e.g., "ATCG" and "GCTA").
    - The range field is disabled for flanking anchors as it uses the sequence between anchors.
4. **Add Reference Sequence (Optional)**:
    - Enter a reference sequence in the **Reference Sequence** field (e.g., "ATCGATCG").
5. **Save the Project**:
    - Click **Save Project**. If all fields are valid, a .json file (e.g., "Experiment_1.json") is created in the specified path.
6. **Switch Language** (Optional):
    - Go to the **Settings** tab to change the interface language.



Note: Depends on the purpose, Barcode, Primer or Anchor sequence can leave empty if you only need to use a specific function.

Note: Anchor is the sequence you believe to be unchanged among the FASTQ files you like to analyze. Use the sequence as close to the editing area as possible.

Note: Flanking Anchor is preferred for most applications.

Sequence Extractor

Overview

The **Sequence Extractor** tab processes FASTQ files to segregate sequences based on barcodes primers defined in the Project Manager. It supports both forward and reverse reads, making it versatile for NGS platforms like Nanopore and Illumina.

Features

- **Extraction Modes**: Extract sequences in both directions (Nanopore) or forward-only (Illumina).
- **File Selection**: Load multiple FASTQ files (.fastq or .fastq.gz) for processing.
- **Barcode Segregation**: Split sequences into files based on barcodes or filter by primer alone.

How to Use

1. **Choose Extraction Mode**:
    - **Extract both forward and reverse** (recommended for Nanopore).
    - **Extract from one direction only** (recommended for Illumina R1 or merged R1+R2 files). To merge R1 + R2 files, see Paired-End Sequence Merge section.
2. **Select Files**:
    - Choose one or more .fastq or .fastq.gz files in the file dialog.
    - The file table will display each file's name and total read count.
3. **Start Extraction**:
    - Click **Step 3: Extract Sequences**.
    - Extracted files are saved in a "Barcode-segregated files" folder within the input file directory (e.g., "sample-[Rep1]-GGAA.fastq").
4. **Monitor Results**:
    - View the extraction report in the text area (e.g., "Barcode GGAA: 68898 sequences, Forward: 36432, Opposite: 32466").

Note: Sequence extraction solely by primer can filter out noise sequence and thus is recommended even if barcoding is not applied in the experiment.

Editing Analyzer

Overview

The **Editing Analyzer** tab analyzes extracted FASTQ files to identify and visualize sequence variants based on set anchor sequence(s). It provides detailed tables and graphical summaries of variant frequencies.

Features

- **Filtering Options**: Set minimum read thresholds and variant display limits.
- **Visualization**: Generate sequence alignment plots or bar charts.
- **Summary**: Aggregate results across groups with exportable CSV files.

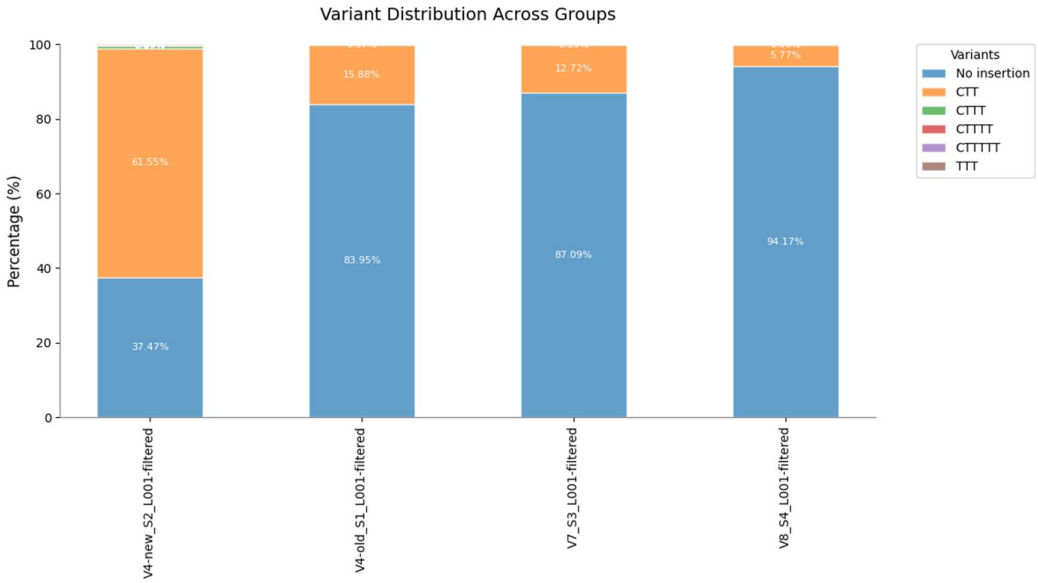
How to Use

1. **Choose File Source**:
    - Under **Step 1**, select:
        - **Use Files from Sequence Extractor** (default, requires prior extraction).
        - **Choose Local Files**: Click **Choose Files** to select .fastq or .fastq.gz files manually.
    - Files appear in the tree widget with editable group names.
2. **Configure Analysis Settings**:
    - **Minimal Reads Threshold**: Select or type a value (e.g., "10") to filter low-frequency variants.
    - **Variants to Display**: Set the maximum number of variants to show (e.g., "10").
3. **Start Analysis**:
    - The progress bar tracks file processing, and results populate the table (e.g., Group, Variant ID, Sequence, Frequency, %).
4. **Summarize Results**:
    - Click **Step 3: Summarize Results** to open a summary dialog.
    - If "**Barcodes for different experimental replicates**" in Project Manager tab was selected, the average percentage of the same variant among groups will be displayed.
    - **Plot Variants**: Select a group or all groups and click **Plot Variants** to view:

- **Bar Plot**: For short ranges (≤3 bp), shows variant distribution for genotyping purposes.
- **Alignment Plot**: For longer ranges, displays sequence alignments.

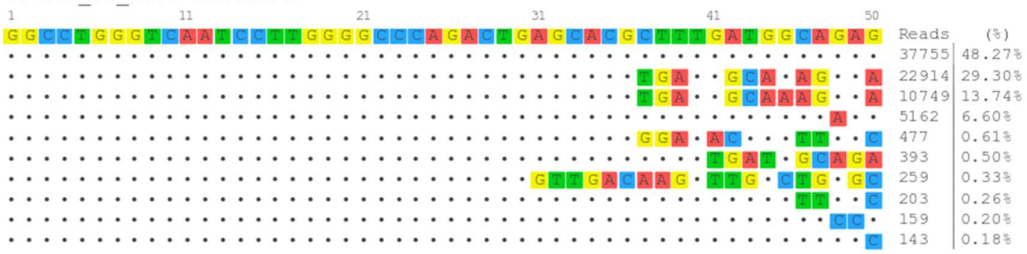
5. **Interpret Results**:
    - Bar plots show stacked percentages; alignment plots highlight differences from the reference or top variant.
    - If reference sequence is not entered in Project Manager tab, the most abundant variant sequence will be used as reference sequence in Alignment Plots.

Example plots:

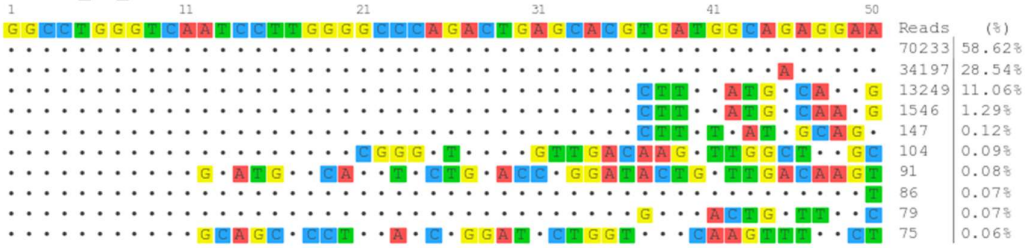


Example plots:

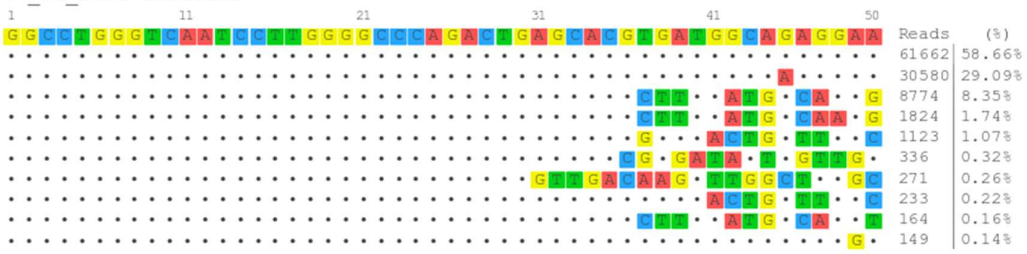
V4-new_S2_L001-filtered



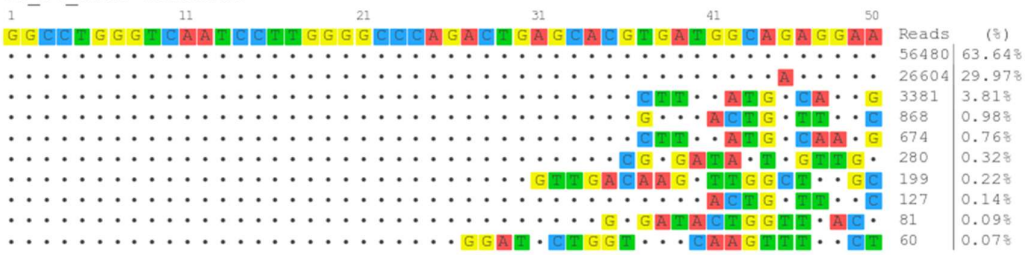
V4-old_S1_L001-filtered



V7_S3_L001-filtered



V8_S4_L001-filtered



Paired-End Sequence Merger

Overview

The **Paired-End Sequence Merger** tab merges paired-end FASTQ files (R1 and R2) into single sequences based on overlapping regions, ideal for Illumina data preparation.

Features

- **Overlap Adjustment**: Set the minimum overlap length for merging.
- **File Pairing**: Automatically group R1 and R2 files by name.

How to Use

1. **Set Minimum Overlap**:
    - Under **Step 1**, choose a minimum overlap length (e.g., "20") from the dropdown.
2. **Select Paired Files**:
    - Click **Step 2: Select paired files**.
    - Choose R1 and R2 FASTQ files (.fastq and .fastq.gz files are both acceptable).
3. **Start Merging**:
    - Click **Step 3: Merge files**.
    - Merged files are saved in a "Merged sequence files" folder (e.g., "sample.fastq").

License Agreement

Upon first launch, you must accept the WEAzer License Agreement. The software is free for personal and research use but prohibits commercial use, redistribution, or modification without permission. See the full license text in the application for details.